

First Semester B.E./B.Tech. Degree Examination, June/July 2024 Mathematics for CSE Stream – I

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	With usual notation prove that $\tan \phi = r \frac{d\theta}{dr}$.	6	L2	CO1
	b.	Find the angle between the curves $r = \frac{a}{1 + \cos \theta}$ and $r = \frac{b}{1 - \cos \theta}$.	7	L2	CO1
	c.	Show that the radius of curvature at any point θ on the cycloid $x = a(\theta + \sin \theta)$, $y = a(1 - \cos \theta)$ is $4a \cos \left(\frac{\theta}{2} \right)$.	7	L3	CO1
OR					
Q.2	a.	Find the pedal equation of the curve $r(1 - \cos \theta) = 2a$.	7	L2	CO1
	b.	Find the radius of curvature for the curve $y^2 = \frac{a^2(a-x)}{x}$ at the point $(a, 0)$.	8	L3	CO1
	c.	Using modern mathematical tool write a program / code to plot the sine and cosine curve.	5	L3	CO5
Module – 2					
Q.3	a.	Expand $\log(\sec x)$ upto the term containing x^4 using Maclausin's series.	6	L2	CO2
	b.	If $u = \log(\tan x + \tan y + \tan z)$, show that $\sin 2x \frac{\partial u}{\partial x} + \sin 2y \frac{\partial u}{\partial y} + \sin 2z \frac{\partial u}{\partial z} = 2$.	7	L2	CO2
	c.	Find the extreme values of the function $f(x, y) = x^2 + y^2 + 6x - 12$.	7	L3	CO2
OR					
Q.4	a.	Evaluate i) $\lim_{x \rightarrow 0} \frac{(a^x + b^x)^{1/x}}{2}$ ii) $\lim_{x \rightarrow 0} \left(\frac{\tan x}{x} \right)^{1/x^2}$.	7	L2	CO2
	b.	If $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$, show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 0$.	8	L2	CO2
	c.	Using modern mathematical tool write a program/code to evaluate $\lim_{x \rightarrow \infty} (1 + 1/x)^x$.	5	L3	CO2

Module – 3

Q.5	a.	Solve : $\frac{dy}{dx} + \frac{y}{x} = y^2x$.	6	L2	CO3
	b.	Find the orthogonal trajectories of $r = a(1 + \cos\theta)$, where a is a parameter.	7	L3	CO3
	c.	Find the general solution of the equation $(px-y)(py+x) = 2p$ by reducing into Clairaut's form by taking the substitution $X = x^2, Y = y^2$.	7	L2	CO3

OR

Q.6	a.	Solve $(y \log y) dx + (x - \log y) dy = 0$.	6	L2	CO3
	b.	Prove that the system of parabolas $y^2 = 4a(x + a)$ is self-orthogonal.	7	L3	CO3
	c.	Solve : $xyp^2 - (x^2 + y^2)p + xy = 0$.	7	L2	CO3

Module – 4

Q.7	a.	i) Find the last digit of 7^{2013} ii) Find the last digit of 13^{37}	6	L2	CO4
	b.	i) Find the remainder when $175 \times 113 \times 53$ is divided by 11. ii) Find the remainder when 2^{23} is divided by 47.	7	L2	CO4
	c.	Encrypt the message STOP using RSA with key (2537, 13) using the prime numbers 43 and 59.	7	L3	CO4

OR

Q.8	a.	Solve $2x + 6y \equiv 1 \pmod{7}$ $4x + 3y \equiv 2 \pmod{7}$.	6	L2	CO4
	b.	Using Fermat's little theorem, show that $8^{30} - 1$ is divisible by 31.	7	L2	CO4
	c.	Show that $4(29)! + 5!$ is divisible by 31.	7	L3	CO4

Module – 5

Q.9	a.	Find the Rank of the matrix $\begin{bmatrix} 2 & -1 & -3 & -1 \\ 1 & 2 & 3 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix}$	6	L2	CO5
	b.	Solve the system of Equations by Gauss-Jordan method $x + y + z = 9$ $2x + y - z = 0$ $2x + 5y + 7z = 52$.	7	L3	CO5
	c.	Using power method, find the largest eigen value and the corresponding eigen vector of the matrix $A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}$. Carry out six iterations.	7	L3	CO5

OR

Q.10	a.	Solve the following system of equations by Gauss-Siedel method. $27x + 6y - z = 85$, $6x + 15y + 2z = 72$, $x + y + 54z = 110$. Carry out three iterations.	7	L3	CO5
	b.	Investigate for what values of λ , μ the equations $x + y + z = 6$, $x + 2y + 3z = 10$, $x + 2y + \lambda z = \mu$ have i) No solution ii) Unique solution iii) Infinite number of solutions.	8	L3	CO5
	c.	Using modern mathematical tool, write a program/code to test the consistency of the equation $x + 2y - z = 1$ $2x + y + 4z = 2$ $3x + 3y + 4z = 1$	5	L3	CO5

GBCS SCHEME

[illegible]

BMATS101

First Semester B.E./B.Tech. Degree Examination, Dec.2023/Jan.2024
Mathematics – I for CSE Stream

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. VTU Formula Hand Book is permitted.

3. *M*: Marks, *L*: Bloom's level, *C*: Course outcomes.

Module – 1			M	L	C
Q.1	a.	With usual notation prove that $\tan \phi = r \cdot \frac{d\theta}{dr}$.	6	L2	CO1
	b.	Find the angle between the curves $r = 6\cos\theta$ and $r = 2(1 + \cos\theta)$.	7	L2	CO1
	c.	Find the radius of curvature for the Folium of De-Cartes $x^3 + y^3 = 3axy$ at the point $\left(\frac{3a}{2}, \frac{3a}{2}\right)$ on it.	7	L2	CO1
OR					
Q.2	a.	Show that the curves $r = a(1 + \sin \theta)$ and $r = a(1 - \sin \theta)$ cut each other orthogonally.	8	L2	CO1
	b.	Find the pedal equation of $r^n = a(1 + \cos n\theta)$.	7	L2	CO1
	c.	Using modern mathematical tool, write a program/code to plot the curve sine and cosine curve.	5	L3	CO5
Module – 2					
Q.3	a.	Using Maclaurin's series, expand $\sqrt{1 + \sin 2x}$ in powers of x upto the terms x^4 .	7	L2	CO1
	b.	If $U = e^{ax+by} f(ax + by)$, prove that $b \frac{\partial U}{\partial x} + a \frac{\partial U}{\partial y} = 2abU$.	6	L2	CO1
	c.	Find the extreme values of the function $\sin x + \sin y + \sin(x + y)$.	7	L3	CO1
OR					
Q.4	a.	Evaluate the $\lim_{x \rightarrow 0} \left[\frac{a^x + b^x + c^x + d^x}{4} \right]^{\frac{1}{x}}$.	8	L2	CO1
	b.	If $U = f(2x - 3y, 3y - 4z, 4z - 2x)$, prove that $\frac{1}{2} \frac{\partial U}{\partial x} + \frac{1}{3} \frac{\partial U}{\partial y} + \frac{1}{4} \frac{\partial U}{\partial z} = 0$.	7	L2	CO1
	c.	Using modern mathematical tool, write a program/code to evaluate $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x} \right)^x$.	5	L3	CO5
Module – 3					
Q.5	a.	Solve $\frac{dy}{dx} + \frac{y}{x} = x^2 y^6$.	6	L2	CO2
	b.	Find the orthogonal trajectories of the family $r = a(1 + \sin \theta)$.	7	L3	CO2
	c.	Find the solution of the equation $x^2(y - Px) = P^2y$ by reducing into Clairaut's form using the substitution $X = x^2, Y = y^2$.	7	L2	CO2
OR					

Q.6	a.	Solve $(8xy - 9y^2)dx + 2(x^2 - 3xy)dy = 0$.	6	L2	CO2
	b.	A voltage Ee^{-at} is applied at $t = 0$ to a circuit of inductance L and resistant R . Find the current at any time t given that the current is initially zero when $t = 0$.	7	L3	CO2
	c.	Solve $x(y')^2 - (2x + 3y)y' + 6y = 0$.	7	L2	CO2
Module - 4					
Q.7	a.	i) Find the last digit in 7^{289} . ii) Find the remainder when $135 \times 74 \times 48$ is divided by 7.	7	L2	CO3
	b.	Solve the linear congruence $6x \equiv 15 \pmod{21}$.	6	L2	CO3
	c.	Using Wilson's theorem, show that $4(29)! + 5!$ is divisible by 31.	7	L2	CO3
OR					
Q.8	a.	Solve the set of simultaneous congruences $x \equiv 5 \pmod{3}$, $x \equiv 2 \pmod{5}$, $x \equiv 1 \pmod{11}$.	7	L2	CO3
	b.	Solve $7x + 3y \equiv 10 \pmod{16}$, $2x + 5y \equiv 9 \pmod{16}$.	6	L2	CO3
	c.	Show that $2^{340} - 1$ is divisible by 31, using Fermat's little theorem.	7	L2	CO3
Module - 5					
Q.9	a.	Find the rank of the matrix $\begin{bmatrix} 1 & 2 & 4 & 3 \\ 2 & 4 & 6 & 8 \\ 4 & 8 & 12 & 16 \\ 1 & 2 & 3 & 4 \end{bmatrix}$	6	L2	CO4
	b.	Solve the system of equation by using Gauss-Jordan method. $x + y + z = 8$, $-x - y + 2z = -4$, $3x + 5y - 7z = -14$	7	L3	CO4
	c.	Using Rayleigh's power method, find the largest eigen value and the corresponding eigen vector of the matrix $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ by taking initial vector as $[1 \ 1 \ 1]^T$. Perform 6 iterations.	7	L3	CO4
OR					
Q.10	a.	Solve the system of equation by using Gauss elimination method. $x + 2y + z = 3$, $2x + 3y + 3z = 10$, $3x - y + 2z = 13$	8	L3	CO4
	b.	Solve the following system of equations by Gauss-Seidal method $20x + y - 2y = 17$, $3x + 20y - z = -18$, $2x - 3y + 20z = 25$	7	L3	CO4
	c.	Using modern mathematical tool, write a programme/code to test the consistency of the equation: $x + 2y - z = 1$, $2x + y + 4z = 2$, $3x + 3y + 4z = 1$	5	L3	CO5

CBCS SCHEME

USN

--	--	--	--	--	--	--	--	--	--

BMATS101

First Semester B.E./B.Tech. Degree Examination, Jan./Feb. 2023 Mathematics – I for Computer Science Engineering Stream

Time: 3 hrs.

Max. Marks: 100

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. VTU Formula Hand Book is permitted.
3. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	With usual notations, prove that $\tan \phi = r \frac{d\theta}{dy}$.	6	L2	CO1
	b.	Find the angle of intersection between the curves $\gamma = \frac{a\theta}{1+\theta}$, $\gamma = \frac{a}{1+\theta^2}$.	7	L2	CO1
	c.	Find radius of curvature of the curve $y = a \log \sec \left(\frac{x}{a} \right)$ at any point (x, y).	7	L2	CO1
OR					
Q.2	a.	With usual notations prove that $\frac{1}{p^2} = \frac{1}{r^2} + \frac{1}{r^4} \left(\frac{dr}{d\theta} \right)^2$.	8	L2	CO1
	b.	Find the radius of the curvature of the curve $r = a(1 + \cos\theta)$.	7	L2	CO1
	c.	Using modern mathematical tool write a program/code to plot the Sine and Cosine curve.	5	L3	CO5
Module – 2					
Q.3	a.	Using Maclaurin's series prove that $\sqrt{1 + \sin 2x} = 1 + x - \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$	6	L2	CO1
	b.	If $Z = e^{ax+by} f(ax-by)$, prove that $b \frac{\partial Z}{\partial x} + a \frac{\partial Z}{\partial y} = 2abz$.	7	L2	CO1
	c.	Find the extreme values of the function $f(x, y) = x^3 + y^3 - 3x - 12y + 20$.	7	L3	CO1
OR					
Q.4	a.	Evaluate $\lim_{x \rightarrow 0} \left(\frac{a^x + b^x + c^x + d^x}{4} \right)^{\frac{1}{x}}$.	8	L2	CO1
	b.	If $u = \frac{2yz}{x}$, $v = \frac{3xz}{y}$, $w = \frac{4xy}{z}$ find $J \left(\frac{u, v, w}{x, y, z} \right)$.	7	L2	CO1

	c.	Using modern mathematical tool write a program code to evaluate $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x$.	5	L3	CO5
Module – 3					
Q.5	a.	Solve $\frac{dy}{dx} + y \tan x = y^3 \sec x$.	6	L2	CO2
	b.	Find orthogonal trajectories of family of curves $r^n = a^n \cos n\theta$.	7	L3	CO2
	c.	Solve $x^2 p^2 + 3xyp + 2y^2 = 0$.	7	L2	CO2
OR					
Q.6	a.	Solve $(x^2 + y^2 + x)dx + xydy = 0$.	6	L2	CO2
	b.	Find the general solution of the equation $(px - y)(py + x) = 2p$ by reducing into Clairaut's form by taking the substitution $X = x^2, Y = y^2$.	7	L2	CO2
	c.	A 12 volts battery is connected to a series circuit in which the inductance is $\frac{1}{2}$ Henry and resistance is 10 ohms. Determine current I, if the initial current is zero.	7	L3	CO2
Module – 4					
Q.7	a.	i) Find the last digit in 13^{37} . ii) Find the remainder when 7^{118} is divided by 10.	6	L2	CO3
	b.	Find the solutions of the linear congruence $12x \equiv 6 \pmod{21}$.	7	L2	CO3
	c.	Find the general solution of linear Dio-phantine equation $70x + 112y = 168$.	7	L2	CO3
OR					
Q.8	a.	Find the remainder when $14!$ is divided by 17.	6	L2	CO3
	b.	Find the solution of system of linear congruences $7x + 3y \equiv 10 \pmod{16}$ $2x + 5y \equiv 9 \pmod{16}$	7	L2	CO3
	c.	Solve $x \equiv 3 \pmod{5}, x \equiv 2 \pmod{6}, x \equiv 4 \pmod{7}$ using Chinese remainder theorem.	7	L3	CO3
Module – 5					
Q.9	a.	Find the rank of matrix $A = \begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$.	6	L2	CO4

	b.	Solve the system of equations by Gauss-Jordan method. $x + y + z = 9$; $2x + y - z = 0$; $2x + 5y + 7z = 52$.	7	L3	CO4
	c.	Find the largest eigen value and the corresponding eigen vector of the matrix $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ taking $[1 \ 1 \ 1]^T$ as initial eigen vector, using power method.	7	L3	CO4
OR					
Q.10	a.	Find the values of λ and μ for which the system $x + y + z = 6$; $x + 2y + 3z = 10$; $x + 2y + \lambda z = \mu$ has i) Unique solution ii) Infinitely many solutions iii) no solution.	8	L2	CO4
	b.	Solve the following system of equations by Gauss-Elimination method $2x + y + 4z = 12$, $4x + 11y - z = 33$, $8x - 3y + 2z = 20$.	7	L3	CO4
	c.	Using modern mathematical tool, write a program/code to test the consistency of the equations $x + 2y - z = 1$, $2x + y + 4z = 2$, $3x + 3y + 4z = 1$.	5	L3	CO5

Model Question Paper-I with effect from 2022(CBCS Scheme)

USN

--	--	--	--	--	--	--	--	--	--

First Semester B.E Degree Examination

Mathematics-I for Computer Science Engineering Stream (BMATS101)

TIME:03Hours

Max.Marks:100

Note:

1. Answer any **FIVE** full questions, choosing atleast **ONE** question from each **MODULE**
2. VTU Formula Hand Book is Permitted
3. M: Marks, L: RBT levels, C: Course outcomes.

Module - 1			M	L	C
Q.1	a	With usual notation prove that $\tan \phi = r \frac{d\theta}{dr}$.	6	L2	CO1
	b	Find the angle between the curves $r = a \log \theta$, $r = \frac{\theta}{\log \theta}$	7	L2	CO1
	c	Show that the radius of curvature at any point of the cycloid $x = a(\theta + \sin \theta)$, $y = a(1 - \cos \theta)$ is $4a \cos\left(\frac{\theta}{2}\right)$.	7	L3	CO1
OR					
Q.2	a	Show that the curves $r = a(1 + \sin \theta)$ and $r = a(1 + \sin \theta)$ cut each other orthogonally.	7	L2	CO1
	b	Find the pedal equation of the curve $\frac{2a}{r} = (1 + \cos \theta)$.	8	L2	CO1
	c	Using modern mathematical tool write a program/code to plot the curve $r = 2 \cos 2\theta $.	5	L3	CO5
Module – 2					
Q.3	a	Expand $\log(\sec x)$ by Maclaurin's series up to the term containing x^4 .	6	L2	CO1
	b	If $u = e^{(ax+by)}f(ax - by)$, prove that $b \frac{\partial u}{\partial x} + a \frac{\partial u}{\partial y} = 2abu$ by using concepts composite functions.	7	L2	CO1
	c	Find the extreme values of the function $f(x, y) = x^3 + 3xy^2 - 3y^2 - 3x^2 + 4$.	7	L3	CO1
OR					
Q.4	a	Evaluate (i) $\lim_{x \rightarrow 0} \left(\frac{a^x + b^x}{2}\right)^{\frac{1}{x}}$. (ii) $\lim_{x \rightarrow 0} \left(\frac{\tan x}{x}\right)^{1/x}$.	7	L2	CO1

	b	If $x + y + z = u, y + z = uv, z = uvw$ find $\frac{\partial(x, y, z)}{\partial(u, v, w)}$.	8	L2	CO1
	c	Using modern mathematical tool write a program/code to show that $u_{xx} + u_{yy} = 0$ given $u = e^x(x \cos(y) - y \sin(y))$.	5	L3	CO5
Module – 3					
Q.5	a	Solve: $\frac{dy}{dx} + \frac{y}{x} = x^2 y^6$.	6	L2	CO2
	b	Find the orthogonal trajectories of $\frac{x^2}{a^2} + \frac{y^2}{b^2 + \lambda} = 1$, where λ is a parameter.	7	L3	CO2
	c	Solve $xyp^2 - (x^2 + y^2)y + xy = 0$.	7	L2	CO2
OR					
Q.6	a	Solve $(x^2 + y^2 + x)dx + xy dy = 0$	6	L2	CO2
	b	When a switch is closed in a circuit containing a battery E, a resistance R and an inductance L, the current i build up at a rate given by $L \frac{di}{dt} + Ri = E$. Find i as a function of t. How long will it be, before the current has reached one-half its final value, if E = 6 volts, R=100 Ohms and L= 0.1 Henry?	7	L3	CO2
	c	Find the general solution of the equation $(px - y)(py + x) = a^2 p$ by reducing into Clairaut's form by taking the substitution $X = x^2, Y = y^2$.	7	L2	CO2
Module – 4					
Q.7	a	Find the least positive values of x such that (i) $71 \equiv x \pmod{8}$ (ii) $78 + x \equiv 3 \pmod{5}$ (iii) $89 \equiv (x + 3) \pmod{4}$	6	L2	CO3
	b	Find the remainder when $(349 \times 74 \times 36)$ is divided by 3.	7	L2	CO3
	c	Solve: $2x + 6y \equiv 1 \pmod{7}$ and $4x + 2y \equiv 2 \pmod{7}$.	7	L3	CO3
OR					
Q.8	a	(i) Find the last digit of 7^{2013} (ii) Find the last digit of 13^{37} .	6	L2	CO3
	b	Find the remainder when the number 2^{1000} is divided by 13.	7	L3	CO3
	c	Find the remainder when $14!$ is divided by 17.	7	L2	CO3
Module – 5					

Q.9	a	Find the rank of the matrix $\begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$	6	L2	CO4
	b	Solve the system of equations by Gauss-Jordan method $x + y + z = 10, 2x - y + 3z = 19, x + 2y + 3z = 22$.	7	L3	CO4
	c	For what values λ and μ the system of equations $2x + 3y + 5z = 9, 7x + 3y - 2z = 8, 2x + 3y + \lambda z = \mu$, has (i) no solution (ii) a unique solution and (iii) infinite number of solutions.	7	L2	CO4
OR					
Q.10	a	Solve the following system of equations by Gauss – Seidel method $10x + y + z = 12, x + 10y + z = 12, x + y + 10z = 12$.	8	L3	CO4
	b	Solve the following system of equations by Gauss-Elimination method $x + y + z = 9, x - 2y + 3z = 8, 2x + y - z = 3$.	7	L3	CO4
	c	Using modern mathematical tool write a program/code to find the largest eigen value of $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$ by power method.	5	L3	CO5

Model Question Paper-II with effect from 2022 (CBCS Scheme)

USN

--	--	--	--	--	--	--	--	--	--

First Semester B.E Degree Examination

Mathematics-I for Computer Science Engineering Stream (BMATS101)

TIME: 03 Hours

Max. Marks: 100

Note:

1. Answer any **FIVE** full questions, choosing atleast **ONE** question from each **MODULE**
2. VTU Formula Hand Book is Permitted
3. M: Marks, L: RBT levels, C: Course outcomes.

	Module - 1			M	L	C
Q.1	a	With usual notation prove that $\frac{1}{p^2} = \frac{1}{r^2} + \frac{1}{r^4} \left(\frac{dr}{d\theta} \right)^2$.	6	L2	CO1	
	b	Find the angle between the curves $r = \frac{a}{1 + \cos \theta}, r = \frac{b}{1 - \cos \theta}$.	7	L2	CO1	
	c	Find the radius of curvature of the curve $y = x^3(x - a)$ at the point (a, 0).	7	L3	CO1	
OR						
Q.2	a	Show that the curves $r = a(1 + \cos \theta)$ and $r = a(1 - \cos \theta)$ Cut each other orthogonally.	7	L2	CO1	
	b	Find the pedal equation of the curve $r(1 - \cos \theta) = 2a$.	8	L2	CO1	
	c	Using modern mathematical tool write a program/code to plot the sine and cosine curve.	5	L3	CO5	
Module – 2						
Q.3	a	Expand $\log(1 + \sin x)$ by Maclaurin's series up to the term containing x^4	6	L2	CO1	
	b	If $u = \log(\tan x + \tan y + \tan z)$, Show that $\sin 2x \frac{\partial u}{\partial x} + \sin 2y \frac{\partial u}{\partial y} + \sin 2z \frac{\partial u}{\partial z} = 2$	7	L2	CO1	
	c	Find the extreme values of the function $f(x, y) = x^2 + y^2 + 6x - 12$.	7	L3	CO1	
OR						
Q.4	a	If $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$ then find the value of $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 0$.	7	L2	CO1	
	b	If $x = r \sin \theta \cos \phi, y = r \sin \theta \sin \phi, z = r \cos \theta$ then find $\frac{\partial(x, y, z)}{\partial(r, \theta, \phi)}$.	8	L3	CO1	

	c	Using modern mathematical tool write a program/code to evaluate $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x$.	5	L3	CO5
Module – 3					
Q.5	a	Solve $\frac{dy}{dx} + \frac{y}{x} = y^2 x$.	6	L2	CO2
	b	Find the orthogonal trajectories of $r = a(1 + \cos\theta)$ where a is parameter.	7	L3	CO2
	c	Solve $p^2 + 2py \cot x - y^2 = 0$.	7	L2	CO2
OR					
Q.6	a	Solve $y(2xy + 1)dx - xdy = 0$	6	L2	CO2
	b	Find the orthogonal trajectories of the family $r^n \sin n\theta = a^n$.	7	L3	CO2
	c	Find the general and singular solution of the equation $(px - y)(py + x) = a^2 p$ reducing into Clairaut's form, using the substitution $X = x^2, Y = y^2$.	7	L2	CO2
Module – 4					
Q.7	a	(i) Find the remainder when 2^{23} is divided by 47 (ii) Find the last digit in 7^{118} .	6	L2	CO3
	b	Find the solutions of the linear congruence $11x \equiv 4 \pmod{25}$.	7	L2	CO3
	c	Encrypt the message <i>STOP</i> using RSA with key (2537, 13) using the prime numbers 43 and 59.	7	L3	CO3
OR					
Q.8	a	Using Fermat's Little Theorem, show that $8^{30} - 1$ is divided by 31.	6	L2	CO3
	b	Solve the system of linear congruence $x \equiv 3 \pmod{5}, y \equiv 2 \pmod{6}, z \equiv 4 \pmod{7}$ using Remainder Theorem.	7	L2	CO3
	c	(i) Find the remainder when $175 \times 113 \times 53$ is divided by 11. (ii) Solve $x^3 + 5x + 1 \equiv 0 \pmod{27}$.	7	L3	CO3
Module – 5					
Q.9	a	Find the rank of the matrix $\begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$	6	L2	CO4
	b	Solve the system of equations by Gauss-Jordan method $x + y + z = 9, 2x + y - z = 0, 2x + 5y + 7z = 52$.	7	L3	CO4
	c	Using Rayleigh's power method find the dominant eigenvalue and the corresponding eigenvector of	7	L3	CO4

		$\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$ by taking $[1 \ 0 \ 0]^T$ as the initial eigenvector [carry out 6 iterations].			
OR					
Q.10	a	Solve the following system of equation by Gauss-Seidel method: $20x + y - 2z = 17, 3x + 20y - z = -18, 2x - 3y + 20z = 25$	8	L3	CO4
	b	Solve the system of equations by Gauss elimination method $2x + y + 4z = 12, 4x + 11y - z = 33, 8x - 3y + 2z = 20.$	7	L3	CO4
	c	Using modern mathematical tool write a program/code to test the consistency of the equations, $x+2y-z=1, 2x+y+4z=2, 3x+3y+4z=1.$	5	L3	CO5

Model Question Paper –I with effect from 2022

USN

--	--	--	--	--	--	--	--	--	--

First Semester B. E Degree examination

Mathematics-I for Computer Science Stream (2MATS11)

Time: 03 Hours

Max. Marks: 100

Note: Answer any **FIVE** full questions, choosing at least **ONE** question from each module.

		Module-1	Marks
Q. 01	a	With usual notation prove that $\tan\phi = r \frac{d\theta}{dr}$	6
	b	Find the angle between the curves $r = a \log\theta$ and $r = \frac{a}{\log\theta}$	7
	c	Show that the radius of curvature at any point θ on the cycloid $x = a(\theta + \sin\theta)$, $y = a(1 - \cos\theta)$ is $4a \cos\left(\frac{\theta}{2}\right)$	7
		OR	
Q. 02	a	Show that the curves $r = a(1 + \sin\theta)$ and $r = a(1 - \sin\theta)$ cuts each other orthogonally	6
	b	Find the pedal equation of the curve $\frac{2a}{r} = (1 + \cos\theta)$	7
	c	Find the radius of curvature for the curve $y^2 = \frac{4a^2(2a-x)}{x}$, where the curve meets the x-axis	7
		Module-2	
Q. 03	a	Expand $\log(\sec x)$ up to the term containing x^4 using Maclaurin's series	6
	b	If $u = e^{ax+by} f(ax - by)$ prove that $b \frac{\partial u}{\partial x} + a \frac{\partial u}{\partial y} = 2abu$ by using the concept of composite functions.	7
	c	Find the extreme values of the function $f(x, y) = x^3 + 3xy^2 - 3y^2 - 3x^2 + 4$	7
		OR	
Q. 04	a	Evaluate i) $\lim_{x \rightarrow 0} \left(\frac{a^x + b^x}{2}\right)^{1/x}$ ii) $\lim_{x \rightarrow 0} \left(\frac{\tan x}{x}\right)^{1/x}$	6
	b	If $u = f(x - y, y - z, z - x)$ show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$	7
	c	If $x + y + z = u$, $y + z = v$ and $z = uvw$, find the values of $\frac{\partial(x,y,z)}{\partial(u,v,w)}$	7
		Module-3	
Q. 05	a	Solve $\frac{dy}{dx} + \frac{y}{x} = x^2 y^6$	6
	b	Find the orthogonal trajectories of $\frac{x^2}{a^2} + \frac{y^2}{b^2 + \lambda} = 1$, where λ is a parameter.	7

	c	Solve $xyp^2 - (x^2 + y^2)p + xy = 0$	7
		OR	
Q. 06	a	Solve $(x^2 + y^2 + x)dx + xydy = 0$	6
	b	When a switch is closed in a circuit containing a battery E, a resistance R and an inductance L, the current i build up at a rate given by $L \frac{di}{dt} + Ri = E$. Find i as a function of t . How long will it be, before the current has reached one-half its final value, if $E = 6$ volts, $R = 100$ ohms and $L = 0.1$ henry?	7
	c	Find the general solution of the equation $(px - y)(py + x) = a^2p$ by reducing into Clairaut's form by taking the substitution $X = x^2, Y = y^2$	7
		Module-4	
Q. 07	a	Find the least positive values of x such that i) $71 \equiv x \pmod{8}$ ii) $78 + x \equiv 3 \pmod{5}$ iii) $89 \equiv (x + 3) \pmod{4}$	6
	b	Find the remainder when $(349 \times 74 \times 36)$ is divided by 3	7
	c	Solve $2x + 6y \equiv 1 \pmod{7}$ $4x + 3y \equiv 2 \pmod{7}$	7
		OR	
Q. 08	a	i) Find the last digit of 7^{2013} ii) Find the last digit of 13^{37}	6
	b	Find the remainder when the number 2^{1000} is divided by 13	7
	c	Find the remainder when $14!$ is divided by 17	7
		Module-5	
Q. 09	a	Find the rank of the matrix $\begin{bmatrix} 2 & -1 & -3 & -1 \\ 1 & 2 & 3 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix}$	6
	b	Solve the system of equations by using the Gauss-Jordan method $x + y + z = 10, 2x - y + 3z = 19, x + 2y + 3z = 22$	7
	c	Using power method find the largest eigenvalue and the corresponding eigenvector of the matrix $A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}$	7
		OR	
Q. 10	a	Solve the following system of equations by Gauss-Seidel method $10x + y + z = 12, x + 10y + z = 12, x + y + 10z = 12$	6
	b	For what values of a and b the system of equation $x + y + z = 6: x + 2y + 3z = 10: x + 2y + az = b$ has i) no solution ii) a unique solution and iii) infinite number of solution	7

	c	Solve the system of equations by Gauss elimination method $x + y + z = 9, x - 2y + 3z = 8, 2x + y - z = 3$	7
--	---	---	---

Table showing the Blooms Taxonomy Level, Course outcome and Program outcome				
Question		Blooms taxonomy level attached	Course outcome	Program outcome
Q.1	a)	L1	CO 01	PO 01
	b)	L2	CO 01	PO 01
	c)	L3	CO 01	PO 02
Q. 2	a)	L1	CO 01	PO 01
	b)	L2	CO 01	PO 01
	c)	L3	CO 01	PO 02
Q. 3	a)	L2	CO 02	PO 01
	b)	L2	CO 02	PO 01
	c)	L3	CO 02	PO 03
Q. 4	a)	L2	CO 02	PO 01
	b)	L2	CO 02	PO 01
	c)	L3	CO 02	PO 02
Q. 5	a)	L2	CO 03	PO 02
	b)	L3	CO 03	PO 03
	c)	L2	CO 03	PO 01
Q. 6	a)	L2	CO 03	PO 02
	b)	L3	CO 03	PO 03
	c)	L2	CO 03	PO 01
Q. 7	a)	L2	CO 04	PO 01
	b)	L2	CO 04	PO 01
	c)	L2	CO 04	PO 02
Q. 8	a)	L2	CO 04	PO 01
	b)	L2	CO 04	PO 01
	c)	L2	CO 04	PO 02
Q. 9	a)	L2	CO 05	PO 01
	b)	L3	CO 05	PO 01
	c)	L3	CO 05	PO 02
Q. 10	a)	L2	CO 05	PO 01
	b)	L3	CO 05	PO 02
	c)	L3	CO 05	PO 01

Bloom's Taxonomy Levels	Lower-order thinking skills		
	Remembering (Knowledge): L ₁	Understanding (Comprehension): L ₂	Applying (Application): L ₃
	Higher-order thinking skills		
	Analyzing (Analysis): L ₄	Valuating (Evaluation): L ₅	Creating (Synthesis): L ₆

Model Question Paper –II with effect from 2022

USN

--	--	--	--	--	--	--	--	--	--

First Semester B. E Degree examination

Mathematics-1 for Computer Science Stream (22MATS11)

Time: 03 Hours

Max. Marks: 100

Note: Answer any **FIVE** full questions, choosing at least **ONE** question from each module.

		Module-1	Marks
Q. 01	a	With usual notation prove that $\frac{1}{p^2} = \frac{1}{r^2} + \frac{1}{r^4} \left(\frac{dr}{d\theta} \right)^2$	6
	b	Find the angle between the curves $r = \frac{a}{1+\cos\theta}$ and $r = \frac{b}{1-\cos\theta}$	7
	c	Find the radius of curvature of the curve $y = x^3(x - a)$ at the point $(a, 0)$	7
		OR	
Q. 02	a	Show that the curves $r = a(1 + \cos \theta)$ and $r = a(1 - \cos \theta)$ cuts each other orthogonally	6
	b	Find the pedal equation of the curve $r(1 - \cos\theta) = 2a$	7
	c	Find the radius of curvature for the curve $y^2 = \frac{a^2(a-x)}{x}$, where the curve meets the x-axis.	7
		Module-2	
Q. 03	a	Expand $\log(1 + \sin x)$ up to the term containing x^4 using Maclaurin's series.	6
	b	If $u = \log(\tan x + \tan y + \tan z)$ show that $\sin 2x \frac{\partial u}{\partial x} + \sin 2y \frac{\partial u}{\partial y} + \sin 2z \frac{\partial u}{\partial z} = 2$.	7
	c	Find the extreme values of the function $f(x, y) = x^2 + y^2 + 6x - 12$.	7
		OR	
Q. 04	a	Evaluate $\lim_{x \rightarrow 0} \left(\frac{a^x + b^x + c^x}{3} \right)^{1/x}$	6
	b	If $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$ show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 0$	7
	c	If $x = r \sin \theta \cos \phi$, $y = r \sin \theta \sin \phi$, $z = r \cos \theta$ find the value of $\frac{\partial(x,y,z)}{\partial(r,\theta,\phi)}$.	7
		Module-3	
Q. 05	a	Solve $\frac{dy}{dx} + \frac{y}{x} = y^2 x$	6
	b	Find the orthogonal trajectories of $r = a(1 + \cos \theta)$ where a is parameter.	7
	c	Solve $p^2 + 2p \cot x - y^2 = 0$.	7
		OR	

Q. 06	a	Solve $y(2xy + 1)dx - xdy = 0$	6
	b	Find the orthogonal trajectories of the family $r^n \sin n\theta = a^n$.	7
	c	Find the general solution of the equation $(px - y)(py + x) = 2p$ by reducing into Clairaut's form by taking the substitution $X = x^2, Y = y^2$	7
		Module-4	
Q. 07	a	(i) Find the remainder when 2^{23} is divided by 47. (ii) Find the last digit in 7^{118} .	6
	b	Find the solutions of the linear congruence $11x \equiv 4 \pmod{25}$.	7
	c	Encrypt the message STOP using RSA with key (2537, 13) using the prime numbers 43 and 59.	7
		OR	
Q. 08	a	Using Fermat's Little Theorem, show that $8^{30} - 1$ is divisible by 31.	6
	b	Solve the system of linear congruence $x \equiv 3 \pmod{5}, \quad y \equiv 2 \pmod{6}, \quad z \equiv 4 \pmod{7}$ using Remainder Theorem.	7
	c	(i) Find the remainder when $175 \times 113 \times 53$ is divided by 11. (ii) Solve $x^3 + 5x + 1 \equiv 0 \pmod{27}$.	7
		Module-5	
Q. 09	a	Find the rank of the matrix $\begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$	6
	b	Solve the system of equations by using Gauss-Jordan method: $x + y + z = 9$ $2x + y - z = 0$ $2x + 5y + 7z = 52$	7
	c	Using power method, find the largest eigenvalue and corresponding eigenvector of the matrix $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$	7
		OR	
Q. 10	a	Solve the following system of equation by Gauss-Seidel method: $20x + y - 2z = 17$ $3x + 20y - z = -18$ $2x - 3y + 20z = 25$	6
	b	Test for consistency $x - 2y + 3z = 2,$ $3x - y + 4z = 4,$ $2x + y - 2z = 5$ and hence solve	7
	c	Solve the system of equations by Gauss elimination method $2x + y + 4z = 12, \quad 4x + 11y - z = 33, \quad 8x - 3y + 2z = 20$	7

Table showing the Blooms Taxonomy Level, Course outcome and Program outcome				
Question		Blooms Taxonomy level attached	Course outcome	Program outcome
Q.1	a)	L1	CO 01	PO 01
	b)	L2	CO 01	PO 01
	c)	L3	CO 01	PO 02
Q. 2	a)	L1	CO 01	PO 01
	b)	L2	CO 01	PO 01
	c)	L3	CO 01	PO 02
Q. 3	a)	L2	CO 02	PO 01
	b)	L2	CO 02	PO 01
	c)	L3	CO 02	PO 03
Q. 4	a)	L2	CO 02	PO 01
	b)	L2	CO 02	PO 01
	c)	L3	CO 02	PO 02
Q. 5	a)	L2	CO 03	PO 02
	b)	L3	CO 03	PO 03
	c)	L2	CO 03	PO 01
Q. 6	a)	L2	CO 03	PO 02
	b)	L3	CO 03	PO 03
	c)	L2	CO 03	PO 01
Q. 7	a)	L2	CO 04	PO 01
	b)	L2	CO 04	PO 01
	c)	L2	CO 04	PO 02
Q. 8	a)	L2	CO 04	PO 01
	b)	L2	CO 04	PO 01
	c)	L2	CO 04	PO 02
Q. 9	a)	L2	CO 05	PO 01
	b)	L3	CO 05	PO 01
	c)	L3	CO 05	PO 02
Q. 10	a)	L2	CO 05	PO 01
	b)	L3	CO 05	PO 02
	c)	L3	CO 05	PO 01

Bloom's Taxonomy Levels	Lower order thinking skills		
	Remembering (knowledge): L ₁	Understanding (Comprehension): L ₂	Applying (Application): L ₃
	Higher-order thinking skills		
	Analyzing (Analysis): L ₄	Valuating (Evaluation): L ₅	Creating (Synthesis): L ₆